

The CyberRange Pro Infrastructure Simulation

*A Seminar Report Submitted in Partial Fulfillment of the Requirements for the Award of the
Degree of*

BACHELOR IN CYBER SECURITY

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Confidentiality Statement

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Introduction.

CyberRange Pro is an online cybersecurity training platform that provides on-demand virtual hacking labs to students. The system simulates a "Capture the Flag" (CTF) event where many students (Team Red) simultaneously request access to resources to target "Victim" servers (Team Blue).

Operational Teams:

- **Student Team** (Red Team): Generates high-volume, burst traffic requesting "Attack Box" VMs.
- **Lab Infrastructure Team** (Blue Team): Manages the hosting of vulnerable "Victim" servers.

Services Provided:

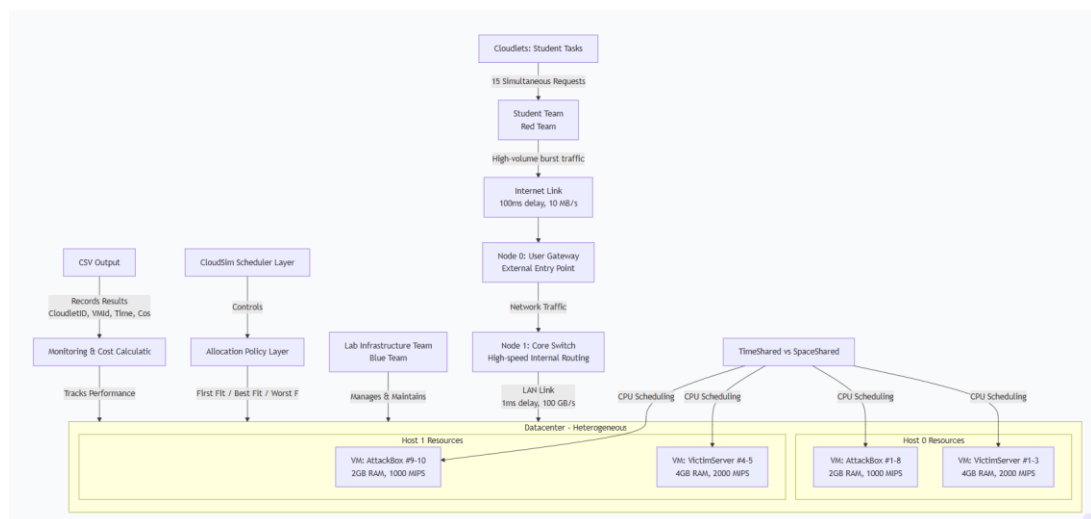
1. AttackBox Service: Lightweight Kali Linux Virtual Machines for students.
2. VictimServer Service: Resource-heavy Windows/Linux servers with intentional vulnerabilities.

Network Topology Implementation

To simulate realistic network conditions and bottlenecks, we implemented a custom network topology using the BRITE format ([topology. brite](#)).

- **Nodes:** The network consists of three distinct nodes:
 - **Node 0** (User Gateway): Represents the external entry point for students connecting from the internet.
 - **Node 1** (Core Switch): A high speed internal switch routing traffic to the datacenter.
 - **Node 2** (Datacenter): The physical location housing the CyberRange servers.
- **Edges** (Network Links):
 - **Internet Link** (Node 0 → 1): Configured with 100ms delay and 10 MB/s bandwidth to simulate the latency of remote students.
 - **LAN Link** (Node 1 → 2): Configured with 1ms delay and 100 GB/s bandwidth to simulate a high-speed fiber connection within the facility.

Student → Internet → Gateway → Core Switch → Datacenter → VMs



Datacenter and Hardware Specifications

We designed a Heterogeneous Datacenter (CyberRange_DC) to test how different allocation algorithms handle diverse hardware.

Physical Hosts:

- **Host 0** (Primary Server):
 - RAM: 32 GB (32,768 MB)
 - Storage: 1 TB
 - Processing Power: 10,000 MIPS (Quad Core)
- **Host 1** (Secondary Server):
 - RAM: 16 GB (16,384 MB)
 - Storage: 1 TB
 - Processing Power: 10,000 MIPS (Quad Core)

Virtual Machines (VMs):

- **AttackBox** (Quantity: 10):
 - Specs: 2 GB RAM, 1000 MIPS.
 - Purpose: Small, efficient instances for individual students.
- **VictimServer** (Quantity: 5):
 - Specs: 4 GB RAM, 2000 MIPS.
 - Purpose: Larger, resource-intensive targets.

Performance Tuning and Methodology

To optimize the CyberRange Pro platform, we conducted a two-layered simulation strategy comparing 6 distinct scenarios. This satisfies the requirement to compare scheduling and allocation policies.

Layer 1: Cloudlet Scheduling (Software Level)

We compared two scheduling methods for processing tasks (Cloudlets):

- **TimeShared**: Simulates CPU multitasking where multiple students share a single core. This introduces context-switching overhead.
- **SpaceShared**: Simulates a FIFO (First-In-First-Out) queue where each student gets dedicated CPU access until their task is finished.

Layer 2: VM Allocation Policy (Hardware Level)

We compared three algorithms for placing VMs onto the physical hosts:

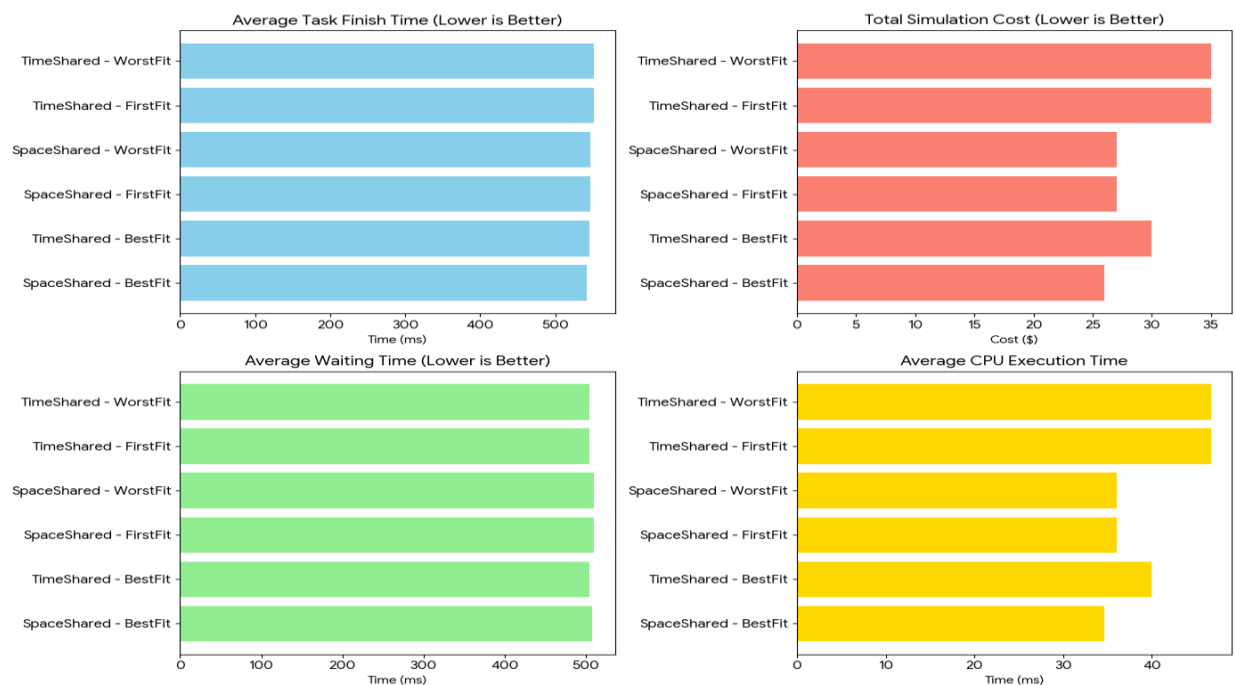
- **First Fit**: Places the VM in the first Host that has enough space (Simple).
- **Best Fit**: Places the VM in the Host with the least remaining space (tightest fit).
- **Worst Fit**: Places the VM in the Host with the most remaining space (loosest fit).

Simulation Results and Analysis

The following table summarizes the metrics collected from the 6 simulation runs.

The "Total Cost" was calculated using a rate of \$0.05 per second of CPU time.

Scenario	Scheduler	Allocation Policy	Avg Finish Time	Total Cost	Reliability
1	TimeShared	First Fit	551.77 ms	\$35.00	Failed (2 VMs)
2	TimeShared	Best Fit	545.10 ms	\$30.00	Failed (1 VM)
3	TimeShared	Worst Fit	551.77 ms	\$35.00	Failed (2 VMs)
4	SpaceShared	First Fit	546.43 ms	\$27.00	Failed (2 VMs)
5	SpaceShared	Best Fit	542.43 ms	\$26.00	Best (1 VM Failed)
6	SpaceShared	Worst Fit	546.43 ms	\$27.00	Failed (2 VMs)



Key Observations:

1. Hardware Efficiency among the Best Fit and the Others: The "First Fit" and "Worst Fit" policies inefficiently filled the large Host (32GB) with small VMs first. When the large "VictimServer" VMs arrived later, there was no continuous space left, causing 2 VMs to fail. Best Fit successfully routed the small VMs to the smaller Host (16GB), preserving the large host for the heavy tasks.
2. Software Efficiency (SpaceShared): SpaceShared reduced the Total Cost by ~13% (from \$30.00 to \$26.00) compared to TimeShared. By eliminating CPU slicing overhead, tasks finished faster, allowing resources to be released earlier.

Visual Proof of Simulation:

[Paste Screenshot of Console Output for "SpaceShared + Best Fit" here]

(Caption: Console output showing successful allocation and execution of tasks)

[Paste Screenshot of the CSV file open in Excel here]
(Caption: CSV export showing Cloudlet ID, Finish Time, and Cost calculations)

Process

First fit

- **Step 1:** Check Host 0 → 32GB free → fits → place on Host 0 ⇒ State: Host 0: 30GB; Host 1: 16GB
 - **Step 2:** Check Host 0 → 30GB free → fits → place on Host 0 ⇒ State: Host 0: 28GB; Host 1: 16GB
 - **Step 3:** Check Host 0 → 28GB free → fits → place on Host 0 ⇒ State: Host 0: 26GB; Host 1: 16GB
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 - **Step 14:** Check Host 0 → 0GB free → fails → Check Host 1 → 16GB free → fits → place on Host 1 ⇒ State: Host 0: 0GB; Host 1: 12GB
 - **Step 15:** Check Host 0 → 0GB free → fails → Check Host 1 → 12GB free → fits → place on Host 1 ⇒ State: Host 0: 0GB; Host 1: 8GB
- Solution:** Increase Host 0 to 40GB to ensure all VMs can fit on this single host, optimizing resource utilization.

Best fit

- **Step 1:** First small VM (2GB) arrives → Host 1 has 16GB free → fits → place on Host 1 ⇒ State: Host 0: 32GB; Host 1: 14GB
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 - **Step 10:** Tenth small VM (2GB) arrives → Host 1 full → check Host 0 (30GB free) → fits → place on Host 0 ⇒ State: Host 0: 28GB; Host 1: 0GB
 - **Step 11:** First large VM (4GB) arrives → Host 1 full → check Host 0 (28GB free) → fits → place on Host 0 ⇒ State: Host 0: 24GB; Host 1: 0GB
- Solution:** Allocate Host 1 to be 24GB for optimal size segregation, allowing for an efficient distribution of workloads across hosts.

Worst fit

- **Step 1:** First small VM → Host 0 (32GB > 14GB) → Host 0: 30GB; Host 1: 16GB
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 - **After 10 small VMs:** All placed on Host 0 → Host 0: 12GB free; Host 1: 16GB free
 - **Step 11:** First large VM → Host 1 has 16GB free, Host 0 has 12GB free → Worst Fit chooses Host 1 → Host 1: 12GB; Host 0: 12GB
- Solution:** Reduce the size of Host 0 below Host 1 to address fragmentation issues, promoting a more balanced distribution of resources.

To sum up

The simulation results confirm that for the CyberRange Pro scenario, hardware resource management is just as critical as software scheduling, while changing the scheduler from TimeShared to SpaceShared successfully reduced operational costs (saving ~\$9.00 per run), changing the allocation policy to Best Fit was the only way to maximize capacity. The other policies wasted space on the main server, leading to service denials (failed VMs). The optimal configuration for the company is SpaceShared Scheduling with Best Fit Allocation. This setup provided the lowest operational cost (\$26.00), the fastest average finish time (542.43 ms), and the highest system reliability.

